

Experiment 3: Questions & Answers

Q1. What is the difference between descriptive statistics and inferential statistics?

- **Descriptive Statistics:** Summarizes, organizes, and describes key features of a dataset using numerical measures (e.g., mean, median, variance) and visual plots without making predictions beyond the data.
- **Inferential Statistics:** Uses sample data to draw conclusions, test hypotheses, and make predictions about a larger population using statistical methods.

Q2. Explain the concepts of the Null Hypothesis (H0) and Alternative Hypothesis (H1).

- **Null Hypothesis (H0):** Assumes that there is no effect, no relationship, or no significant difference between variables in a population.
- **Alternative Hypothesis (H1):** Assumes that there is a statistically significant effect, relationship, or difference between variables.

Q3. What is a p-value? How is it used to make statistical decisions?

- **Definition:** The probability of obtaining test results at least as extreme as the observed results, assuming the null hypothesis (H0) is true.
- **Decision Rule:**
 - **$p \leq 0.05$:** Reject H0 (statistically significant).
 - **$p > 0.05$:** Fail to reject H0 (insufficient evidence).

Q4. Differentiate between a t-test and ANOVA. In which situations is each test applied?

- **Independent Sample t-test:** Compares the means of **2 independent groups** (e.g., comparing income between employees who leave vs stay).
- **One-Way ANOVA:** Compares the means across **3 or more independent groups** simultaneously (e.g., comparing job satisfaction across multiple job roles).

Q5. What does the Pearson correlation coefficient indicate? What are its possible values?

- **Indication:** Measures the direction and strength of a linear relationship between two continuous variables.
- **Range:** Values range from **-1 to +1**:
 - **+1:** Perfect positive linear correlation.
 - **0:** No linear correlation.
 - **-1:** Perfect negative linear correlation.

Q6. Explain the significance of R² (Coefficient of Determination) in Linear Regression.

- **Significance:** R² represents the proportion of variance in the dependent variable explained by the independent variable(s).

- **Interpretation:** Values range from 0 to 1 (or 0% to 100%). An R^2 of 0.85 means 85% of the outcome's variability is explained by the regression model.

Q7. Why is statistical analysis important before applying machine learning algorithms?

- Detects anomalies, missing values, and potential outliers in the data.
- Identifies multicollinearity between features to avoid redundant inputs.
- Helps select relevant features based on statistical significance.
- Validates key assumptions (such as normality and variance homogeneity) needed for statistical ML models.

Q8. What assumptions should be satisfied before performing a t-test or ANOVA?

- **Normality:** Data in each group follows an approximately normal distribution.
- **Homogeneity of Variance:** Variances across groups are approximately equal.
- **Independence:** Samples and observations are independent of each other.
- **Continuous Scale:** The target variable is measured on an interval or ratio scale.

Q9. How can regression analysis help organizations in forecasting and decision-making?

- **Forecasting:** Predicts continuous outcomes (e.g., estimating salary based on experience or forecasting sales).
- **Impact Measurement:** Quantifies how changes in independent variables influence business target metrics.

Q10. Give two real-world applications where hypothesis testing is commonly used in data analytics.

1. **A/B Testing in Digital Marketing:** Testing whether a new webpage design increases user conversions compared to the original design.
2. **Healthcare & Clinical Testing:** Testing whether a new medical treatment significantly improves patient recovery rate compared to a control group.